

Characterization and optimization of a tapered amplifier by its spectra through a long multi-pass rubidium absorption cell

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We present a method to characterize and optimize a tapered-amplifier laser system (TALS) by its spectral quality through a long multi-pass rubidium absorption cell. A thermal vapor cell is used to measure the non-resonant spectrum of TALS, including the broadband amplified spontaneous emission (ASE), which is its main spectral noise. This method gives a simple quantified measurement to optimize various working parameters of a TA, including current and temperature online. It can as well be used to compare various TA chips based on their usage time during our precision measurement experiments. The results of this method are compared and found in sync with traditional methods of Fabry–Perot cavity and beat measurements. Such characterization and optimization are important for noise control in atom interferometers, atomic clocks, and other atomic manipulations. It can very well be used for investigating nonlinearity and ASE inside amplifying chips and can be utilized in other applications of ASE using bioimaging. © 2018 Optical Society of America

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1. INTRODUCTION

Tapered-amplifier laser systems (TALSs), with a typical gain of 20–30 dB and a sideband suppression ratio of 40 dB, form the core of many modern atomic physics and quantum optics experiments, as well as in communication wherein high powers are required. These include precision measurements using atom interferometry [1–3] or far-off-resonance dipole traps [4], or regular laser cooling experiments where various frequency beams are needed for cooling, repumping, and other manipulations [5]. Typically, these amplifiers have high gain, mirrorless laser type structure, small active volume because of slender structure of the chip, and small recombination lifetimes and hence wide linewidths compared to most other lasers. However, these properties make them generate some unwanted noise due mainly to the spontaneous emission processes called amplified spontaneous emission (ASE) [6]. Its origin can be easily understood by modeling the optical amplifier as a two-level quantum system [7]. The material gain coefficient as calculated from such a model for an optical amplifier is given by the expression

$$g_m = \frac{A_{21} c^2 l(v) (N_2 - N_1)}{8 \pi n_r^2 v^2}, \quad (1)$$

where two levels of energies E_1 and E_2 ($E_2 > E_1$), respectively, with population N_1 and N_2 , have frequency related to the spontaneous emission as ν . Here, A_{21} is the spontaneous emission parameter from 2 to 1, c the velocity of light, $l(v)$ the normalized transition line-shape function for the transition between level 1 and 2, and n_r the material refractive index. One can draw two conclusions from the above equation. First, to achieve gain in the optical amplifier, population inversion ($N_2 > N_1$) should exist between two levels, which is the basic requirement of a laser amplifier. Second, due to the presence of A_{21} in the expression, there will be spontaneous emission (or noise) and hence ASE accompanying the resulting gain. As the process underlying ASE generation viz. spontaneous emission is random in nature, it is highly incoherent in the time domain, thus broadband, though it preserves the spatial coherence behavior [6]. Another type of noise in TAs is due to the beat-note between signal and spontaneous emission components [8,9]. Apart from this, in an older TA chip, the cavity

effect due to linear cavity formation caused by unwanted reflections [6] converts a traveling wave amplifier to a Fabry–Perot (FP)-type amplifier [7]. This conversion causes ripples in the gain and hence a certain amount of noise at the output of the amplifier. Yet another source of noise is cross talk due to the nonlinear four-wave mixing [7]. All these noise components do affect the precision involved in the before-mentioned experiments. In particular, in the case of an atom interferometer, such noise from the Raman-beam-generating amplifier causes ac-Stark shift, which is sufficient to affect the resulting precision [10]. It can cause de-coherence in the Bloch oscillations due to broadband spontaneous emission along with the resonant light [11]. Further, it affects accuracy in atomic lattice clocks [12–14] and causes parametric heating of atoms in an atomic lattice [15]. In communication, such noise give rise to cross talk between multiple channels in wavelength-division-multiplexed systems [7]. Thus, to improve the measurement accuracy in all these applications, one needs to carefully understand and quantify the noise from such devices. As these noise components partly arise from the nonlinearity inside the TA chip, such measurements can be used to understand it further, including the phenomenon of four-wave mixing and terahertz generation [16].

The present work is an attempt to give a method to measure the noise at the output of the TALs. As explained above, ASE being the main part of the noise, our focus is on it. It has been studied well in the case of a gas laser by Allen and Peters in Ref. [17] and subsequent work in detail. However, a good measurement in the case of a TA in and around the mid-infrared (IR) range is needed due to its importance in laser cooling of alkalis and ionic species. Further, this wavelength is important as well in communication due to its proximity to the second harmonics of the C band. Apart from this, due to its unique temporal and spatial coherence, ASE has many applications in

varied fields, such as bioimaging using optical coherent tomography [18] and far-off-resonance atom traps [19]. There have been attempts to understand the coherence properties of ASE experimentally using $g^{(2)}$ correlation measurement [20] and theoretically by using second quantized theory [21]. Thus, in this regard, the present instrument and the method developed here give another good parameterization of ASE for further study and applications in similar fields.

2. PRINCIPLE OF MEASUREMENT AND EXPERIMENTAL DETAILS

Work by Liu *et al.* [22] had concluded that the output of such semiconductor systems can be described as a combination of a coherent, highly stabilized, sinusoidal oscillation in addition to an additive Gaussian noise representing the net effect of spontaneous emission from the device. Based on this conclusion, we use a multi-pass rubidium (Rb) vapor cell to investigate and characterize a TA. These cells have already been used in absorption mode in atmospheric studies as well as in analytical chemistry using the photo-acoustic method [23]. One main advantage of using these cells is either they are readily available in various atomic physics laboratories or if not, they are easy to design. In the present work, we use our cell as a filter for the resonant light to analyze the noise output of the TA chip. For this, the cell is put in the path of a frequency-locked diode laser beam. The diode is locked to a resonance transition of Rb. The locked resonant beam when passing through the cell gets absorbed. Thus, the cell effectively cuts the coherent sinusoidal oscillation part as mentioned above. This leaves the non-resonant, noisy part of the beam. This is detected by the photodiode and analyzed to get various conclusions in the present work.

Our experimental schematic is as shown in Fig. 1. A 30 mW beam from a New Focus diode laser lasing at 780 nm is used.

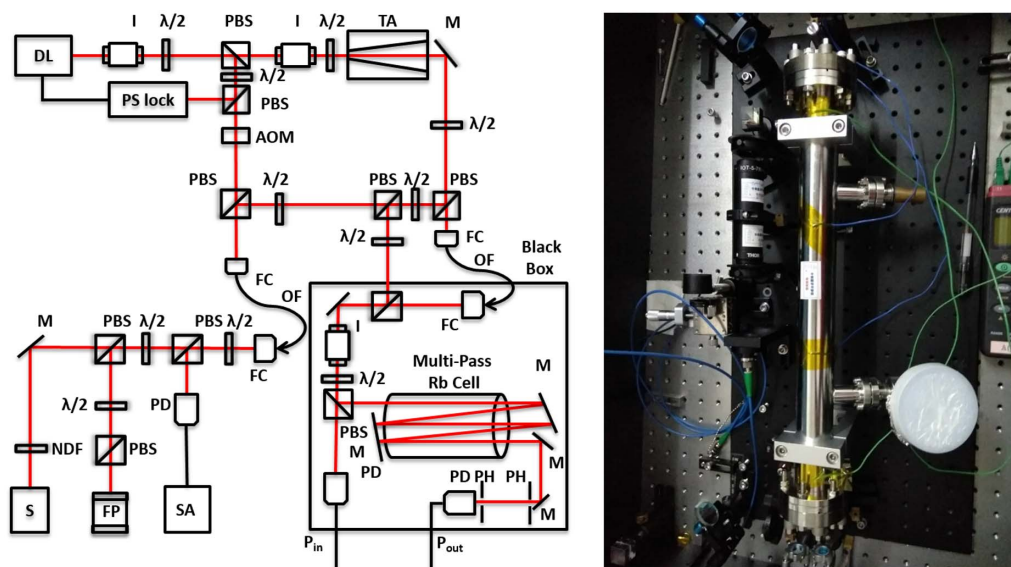


Fig. 1. Experimental schematic for residue measurement measured with our multi-pass Rb cell and compared with spectrometer (S), Fabry–Perot cavity (FP), and spectrum analyzer (SA). DL, diode laser; PS, polarization spectroscopy; I, isolator; $\lambda/2$, half-wave plate; PBS, polarization beam splitter; AOM, acousto-optic-modulator; M, mirror; NDF, neutral density filter; FC, fiber coupler; OF, optical fiber; TA, tapered amplifier; PH, pinhole. Photo on the right shows the multi-pass cell.

Similar to our previous work [24], to narrow down the line-width of the laser, it is locked to one of the four manifolds of the Rb D_2 transitions using polarization spectroscopy [25]. Part of this locked beam is shifted in frequency by an acousto optic modulator (AOM) at 80 MHz for heterodyne beat measurement later. The un-shifted beam is sent through the TA (m2k-laser-GmbH or Eagleyard Photonics) for amplification. Output of the TA is then partly sent through the multi-pass cell for the measurement. The other part of the beam is sent to the comparing devices. These include a spectrometer (Ocean Optics, QE65-pro), a FP cavity (FPI801, UniQuanta) for mode evaluations and any sideband detection, and a high-speed photo detector along with the shifted AOM beam for heterodyne beat measurement. Laser powers are controlled in various parts of the experiment using regular half-wave-plate-PBS (polarizing beam splitter) combinations and with a neutral density filter (NDF) for the spectrometer. Fifty-five dB isolators from ThorLabs are used at appropriate places to avoid any feedback destabilization or any accidental damage in extreme cases to the laser system and TA.

Si photodiodes are used for measuring the P_{in} (input power) and P_{out} (output power) for the measurement cell. The half-wave-plate-PBS combination before the input port of the cell is duly balanced so as to appropriately measure the P_{in} at the photodiode. A double pinhole (PH) assembly is used before the output photodiode to avoid any stray light entering the detector. A multi-pass cell (Fig. 1) is a 40 cm long stainless steel (SS) vacuum pipe with anti reflection-coated K9 windows from Union Optics. It is evacuated to $\sim 10^{-4}$ Pa using a turbo-molecular pump backed with a dry rotary pump. Temperature for the cell is monitored using thermocouples. The multi-pass cell along with the detection system is put inside a black box to avoid any background light to enter the measurement detectors used for the input and output power measurements. The parameter measured in the experiment is the residual power after the cell, which we call residue (η) defined as

$$\eta = \frac{(P_{out} - P_{stray})}{(P_{in} - P_{stray})}, \quad (2)$$

where P_{stray} is the stray light falling on the detectors with the laser beam blocked. Due to the black box used, the typical P_{stray} value we measured is as low as ~ 5 nW. As is well known, the absorption through the vapor cell is governed by Beer–Lamberts law [6]:

$$I = I_0 e^{-\alpha L}, \quad (3)$$

where I_0 is the input intensity to the cell and I is the output intensity after it, α is the absorption coefficient, and L is cell length. Our experiment demands a higher level of absorption so that the resonant part of the beam gets cut as much as possible. We choose ^{85}Rb $F = 3 \rightarrow F' = 4$, the cooling transition to lock the laser. We find that with our experimental conditions, five passes through the cell suffice for getting almost 100% absorption for a wide range of input power from 4 μW to 2 mW [Fig. 2(a)]. The absorption saturation seen in the figure is in agreement with the work by Sagel *et al.* [26]. In their work on cesium (Cs) absorption, they used a rate equation for the atomic species modeled as a three/multi-level atomic system. They considered alignment of population in the presence of trapping levels, transit relaxations, and magnetic field mixing. However, the corner (or knee) power of the saturation is at a higher value in our case. This is due to the higher absorption cross section for ^{133}Cs ($346.9 \times 10^{-15} \text{ m}^2$) compared to ^{85}Rb ($290.7 \times 10^{-15} \text{ m}^2$) [27], which makes the saturation happen at a lower value for Cs as compared to Rb. Next, as is seen in the data, saturation of the transition happens at a higher power when we increase the number of passes through the cell. This is due to increase in the number of atoms participating in the absorption. Next, for choosing the locking transition, we measure the residue with the laser locked to four possible transitions in Rb D_2 line corresponding to the four ground hyperfine levels for the two isotopes. Variation of the residue with input power for different lock points used is shown in Fig. 2(b). As is evident, transitions start getting saturated after a certain input power starting from two repump transitions, ^{87}Rb $F = 1 \rightarrow F' = 2$ at around 32 μW , ^{85}Rb $F = 2 \rightarrow F' = 3$ at 256 μW ; for the two cooling transitions, ^{87}Rb $F = 2 \rightarrow F' = 3$ at 512 μW , ^{85}Rb $F = 3 \rightarrow F' = 4$ at 2048 μW . This is because of the different isotopic abundances of the two isotopes

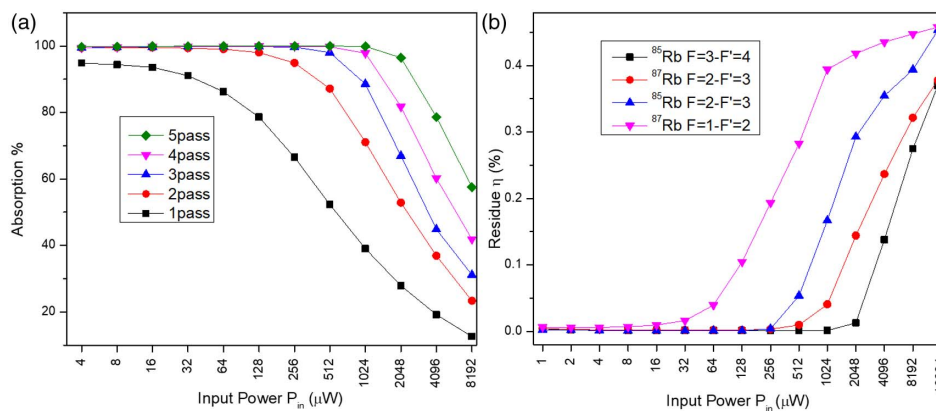


Fig. 2. Optimizing the number of passes through the cell (a) and the lock point (b). (a) Absorption variation with P_{in} for different numbers of passes through the vapor cell. Corner power (see text) is the power when absorption saturates. (b) Residue variation with the input power for different choices of lock points. Frequency dependence of residue after certain input power is due to the effect of unabsorbed resonant light.

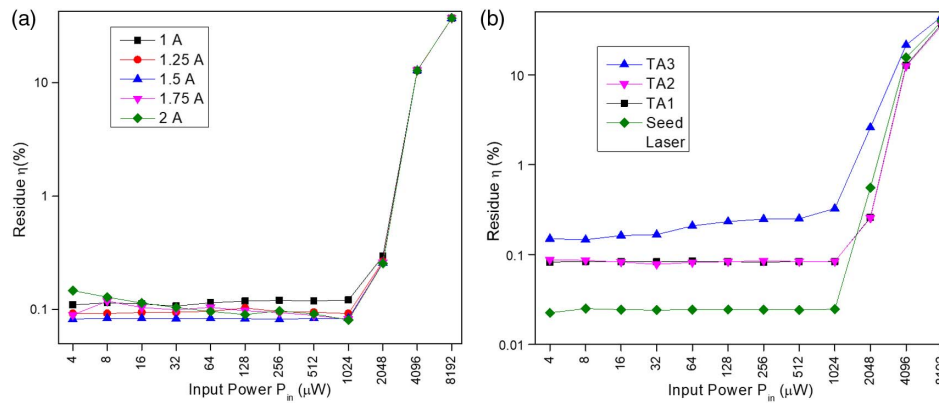


Fig. 3. Residue variation with the input power for (a) a brand new TA chip TA1 with different TA currents. It is seen that for TA1, residue shows little fluctuation for different current values, making it suitable as a standard for comparing with other TAs. (b) Comparison of various TAs with different TA chips TA1, TA2, and TA3.

(^{85}Rb : ^{87}Rb is around 72:28) along with their different transition strengths [27]. So, we choose ^{85}Rb cooling transition for our measurement due to its maximum dynamic range prior to the start of saturation where the frequency effect comes in to affect the residue. These frequency effects come because of the unabsorbed resonant light by the vapor cell. Once we choose these parameters for our experiment, we measure the residue for different TA chips for different input powers to the vapor cell (Fig. 3) with the experimental setup using the method described below.

One important observation can be made in Fig. 2(a). The constant absorption of the cell despite environmental fluctuation in its temperature or any other parameter for a wide range of input power from 4 μ W until almost 2 mW in the case of five passes is important. Due to this, we can easily choose the residue as a standardization and optimization parameter to compare various TAs, and use it to optimize them as well for a wide dynamic range of input powers.

3. EXPERIMENTAL MEASUREMENT AND RESULTS

We first lock the diode laser to the cooling transition line ($F = 3 \rightarrow F' = 4$) of ^{85}Rb using polarization spectroscopy. This laser is used as a seed and injected into the TA to be tested. We take a brand new TA, labeled as TA1 as standard and compare all the TA residue measurements with respect to this. Residue measurement for TA1 is shown in Fig. 3(a). As is evident, there is very small variation of residue value around 0.1 with change in current varying from 1 A to 2 A. This makes it more logical to use this TA as a standard for our further measurements. Now we test various TALSs with different TA chips from different companies and with different usage times taking TA1 as standard. Power of the seed laser used is 20 mW for the TA chip at 22.4°C with TA current as 1.5 A. The result is shown in Fig. 3(b). We can see that the seed laser shows the minimum residue, thus emphasizing the absence or very low level of noise related to the spontaneous amplification or any other type of noise from the locked diode laser. This is because the laser used is in external cavity (ECDL)

configuration, which narrows its linewidth by using the grating feedback. This is further reduced by using electronic locking with polarization spectroscopy. Next, the difference in the residue of different TA chips is attributed to the usage they have gone through. Accidental damage to some of them causes the anti-reflection coating to be damaged, thus showing higher residue. However, there is a common threshold for an unabsorbed resonant beam around and above 1 mW input power. This is because of power saturation and thus increases in the unabsorbed part of the laser beam. So, the true noise is measured below this threshold level. One important aspect of our method unlike other traditional methods of spectral characterization, such as cavity or beat measurement, is that we can optimize the parameters of the TA online or while working on it. This is because of a one-to-one relation between the residue measured and TA parameters. To show this optimization process experimentally, we use residue measurement and optimize TA4 using our method for choosing the best working conditions with regard to current, temperature, and the input seed power. The results are shown in Fig. 4. Figure 4(a) shows the residue variation with TA current at different seed powers; Fig. 4(b) shows the same variation for different temperatures. The TA chip temperature in Fig. 4(a) is 22.4°C. It is evident from the data that initially with increasing the TA current from 1 A until around 1.2 A, the residue decreases. This is because, as the current increases, the pumping rate of the population inversion also increases, and hence the increase in stimulated emission as compared to the spontaneous emission. This results in the decrease of residue [7]. However, with further increase in current, as the population inversion increases, the average number of atoms per unit volume N_2 in the excited level also increases. This makes the spontaneous emission also higher along with the TA current, which results in higher residue as well. This trend is clearly shown in our data.

For the seed power, we find that the higher the seed power the better the working of the TA. This is due to the frequency pulling of the TA with the seed acting as a master laser and the TA chip being the slave. With higher seed power, this effect becomes more prominent and results in lowering of residue with increase in seed power. Next in Fig. 4(b), we measure

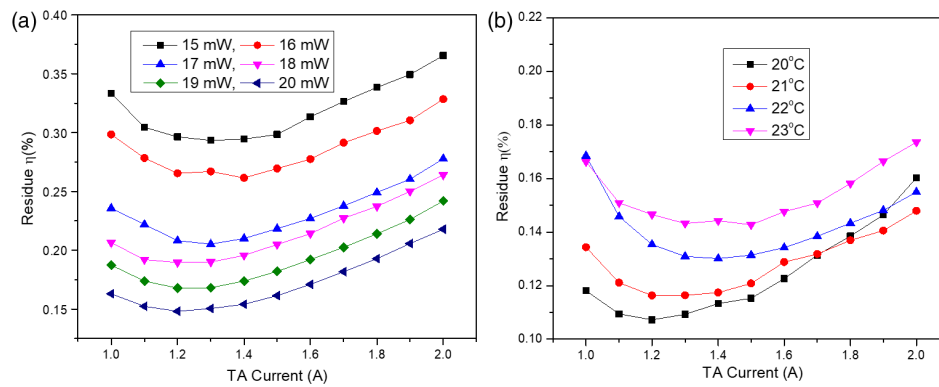


Fig. 4. TA optimization using residue measurement versus TA current for different (a) seed powers and (b) temperatures.

the effect of temperature of the TA chip on the residue. We found that with lower temperature, the residue is found to be lower. It is due to the fact that with lower temperature, N_2 remains low, and hence the spontaneous emission, and thus the residue, remains low. So, the optimum parameters found for the TA chip being tested are as follows. TA current should be in the range of 1.2–1.4 A. Seed power should be as high as possible with maximum value as given in the TA specifications. TA chip temperature should be as low as possible, again capped by its specification values. In our present scenario, we found the best seed power to be 20 mW and TA temperature to be 20°C.

As we know, absorption is affected by number density of atoms in the atomic vapor, magnetic inhomogeneity, possible collision effects, as well as transit time broadening. As our method takes resonant light absorption as the basis, we discuss the effect of these aspects one by one on our measurements now. A simple increase in number density of atoms increases the absorption of resonant light. This will result in increase of the saturation point for the residue value and will give a higher dynamic range to measure residue in our method. This fact is clearly visible in Fig. 5 where we have done the

measurement of residue for different temperature of the cell. This results in increase in number density of absorbing atoms. We can clearly see as the temperature of the cell increases, the flat region of the graph increases as well. Thus, we get a higher dynamic range to measure residue as we increase the temperature. Next, the cell used for locking the laser is well shielded by the magnetic shielding, so it will not have any effect on our experiment. As the length of the measurement cell is only 40 cm, we do not expect any drastic variation in magnetic field resulting in any magnetic inhomogeneity around it. We measured the magnetic field around the cell and found the variation within a maximum of ~ 0.05 G. As the Zeeman shift for a rubidium atom is ~ 1 MHz/Gauss [27], the magnetic field inhomogeneity will cause shift in the absorption profile at the level of ~ 50 kHz, which is miniscule for our result, as we are dealing with a very high absorption rate by using multi-passes through the cell. Thus, it will not alter or affect our conclusions regarding TA. Next are the collision effects inside the atomic cell among the Rb atoms. For self-collisions, the order of magnitude of shift/broadening is $\sim$ kHz [23]. This is again very small as compared to the linewidth of the laser, so it does not affect our conclusions. For the transit time broadening, we need to consider the transit time a typical atom takes (with average velocity of 300 m/s) to pass through the laser beam of size ~ 2 mm. As it will be on the order of 3 ms and the lifetime of the Rb transition is $\sim 10^{-9}$ s, it is evident that the atom will go through a lot of decay cycles before transiting through the laser beam. Hence, the effect of this broadening will average out and will not affect our results.

4. VERIFICATION AND COMPARISON OF OUR RESULTS WITH OTHER METHODS

Finally, we compare our residue measurement conclusions with other techniques commonly used for spectral characterization. We choose a spectrometer, a FP cavity, and a heterodyne beat measurement. All the measurements done in this section are done with an older chip so as to have higher residue variation from 1% until 17%.

First, we take a spectrometer (QE65 from Ocean Optics) whose resolution is quoted as 0.1 nm. Part of the comparison beam is taken through a NDF for controlling the incident power to the detector. Around ~ 1 μ W of input beam is used

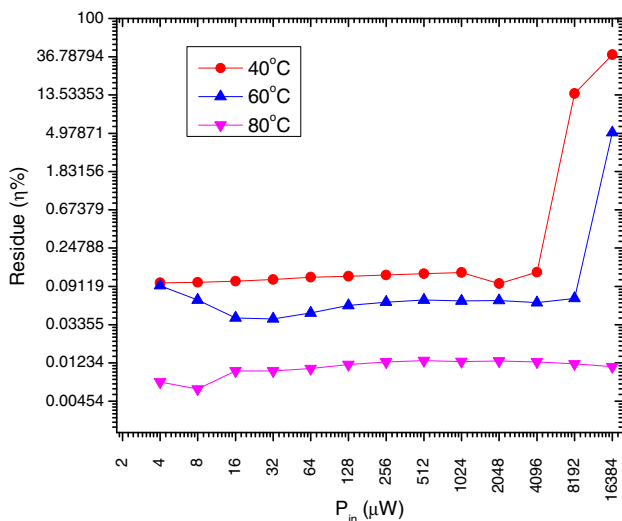


Fig. 5. Temperature variation of residue showing the increase of dynamic range of our measurement as the temperature, and hence the absorption due to the cell, increases.

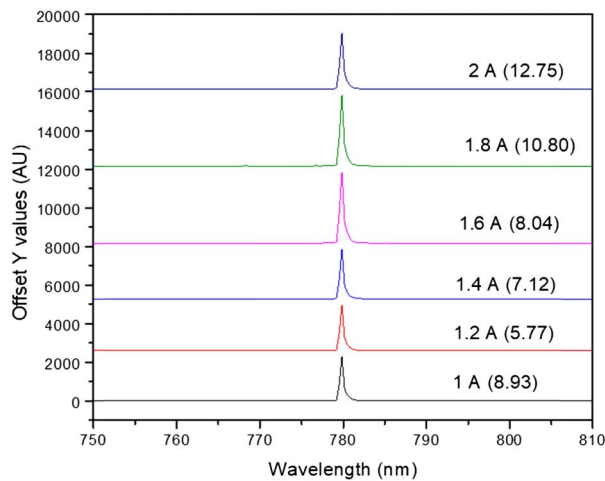


Fig. 6. Spectrometer data for different TA currents (currents increasing from bottom to top). Due to low resolution (0.1 nm), there are no sidebands seen, though the residue (shown in the brackets in percentage along with graphs) changes quite a lot. There is small variation in width and height of peaks, but as the data points are not sufficient to analyze, no proper conclusion can be drawn.

here so as not to saturate the detector. Broadband ASE is expected to show some sidebands in the spectrum. However, due to the low resolution (Fig. 6), we could not see much difference in the spectra recorded as the TA current is varied from 1 A to 2 A, though the residue measured by our method varied a lot (residue is shown in brackets along with the graphs). There are no sidebands seen as the wavelength is scanned across the resonance. The only difference we found is slight variation in the width and the height of the peak shown at the resonant wavelength, i.e., 780 nm. This means that the noise sidebands are all embedded in the resonance peak and are not visible due to the low resolution of the instrument. We could not quantify the change in the width and height of the spectrum because of the low number of the data set, which makes it impossible to have a fit with good value of fit-residuals. Probably it needs a higher resolution grating spectrometer for resolving the sidebands, which is though not a very commonly found instrument in a laboratory due to its higher cost.

Second, we use a confocal FP cavity with a free spectral range (FSR) of ~ 1.5 GHz and finesse ~ 200 . Part of the comparison beam is aligned with the linear cavity with input power of ~ 1.5 μ W. The temperature used for the TA chip is 22.4°C. Seed power used is 20 mW, and the power input to the cell is 100 μ W. A typical set of cavity modes with the measured values of residues is shown in Fig. 6(a). We can see as the residue increases from 1% to 15%, the cavity modes change in the height along with the emergence of sidebands. However, such changes as well as sideband height and positions are found to be totally random in nature. This is because the causal spontaneous noise is statistically stationary and result in fluctuation in both amplitude and phase of the output optical signal [9]. So, though the cavity shows the existence of the noise in the TA output, it is very difficult to quantify it using sidebands with such randomness. In a regular scenario of spectral characterization with a FP cavity, periodic-mode-generating-light sources are used. Stable sidebands get generated at periodic known intervals in such cases [5]. So, it becomes easier to analyze and get quantitative conclusions from them. However, it is not the case in our experimental scenario. Another way of using the cavity for this purpose is in reflection mode, where it works as a narrow bandwidth absorber. However, we found the absorption to be of low level in the case of our cavity. These result in modes that cannot give any conclusive answer about the spectral nature of TA output. Instead, we try to correlate the height of the cavity modes [Fig. 7(b)] with the increasing residue. These heights are found indirectly by using an Origin software routine. It shows the general trend but still fluctuates for higher values of residue. Similar results are found in the case of a 300 MHz FSR cavity, which we used for looking for any sidebands generated near the vicinity of the main cavity modes.

Finally, we use heterodyne beat note measurement between the frequency-shifted seed beam by using an 80 MHz AOM and another un-shifted one at TA output. It is detected by a high-speed detector (New Focus, model 1431) and recorded using a 13.6 GHz spectrum analyzer. Power used on the detector from the two beams is 100 μ W each. Figure 8(a) shows data for a scan range of 1 GHz to look for any sidebands apart from the expected 80 MHz beat signal. The frequency peak and the sidebands seen in the figure are the beat signals at ~ 80 MHz and the higher harmonics of the same appearing

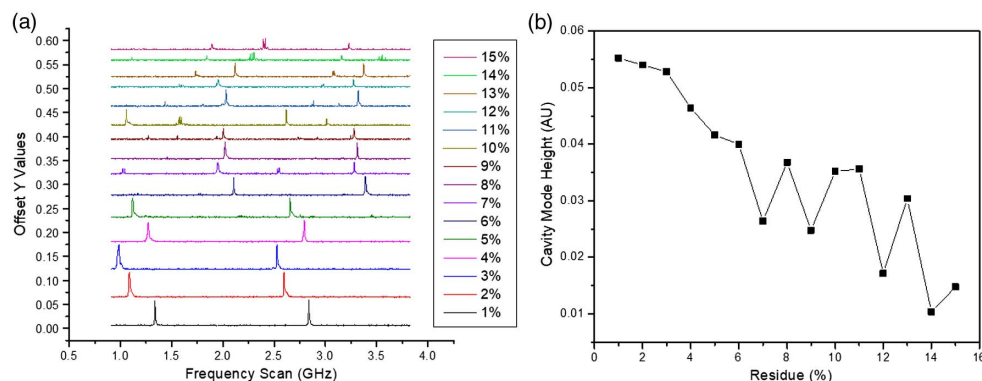


Fig. 7. (a) Cavity modes for different values of measured residues (in percentage shown with curves with residue increasing from bottom to top from 1% to 15%). (b) Peak height of the cavity mode along with the measured residue.

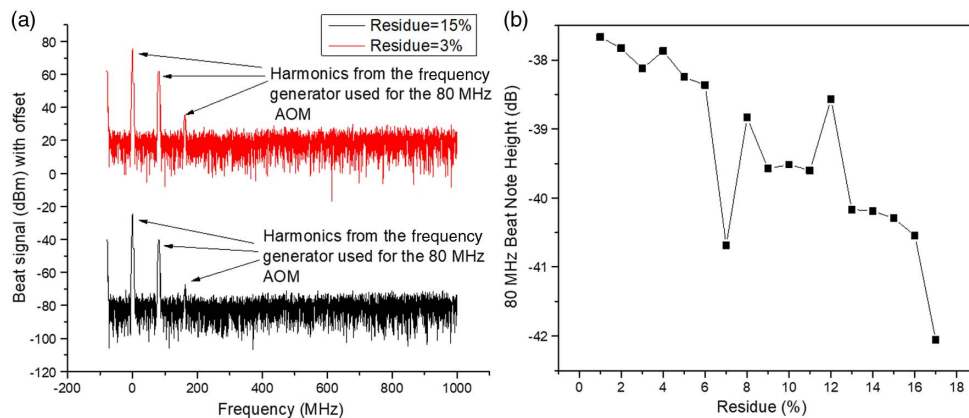


Fig. 8. (a) Beat note as measured for 3% (upper trace) and 15% (lower trace) residue with a 1 GHz scan range. Curves show beat note at 80 MHz along with the electronics sidebands due to the higher harmonics of the AOM driver. However no sidebands are visible due to any noise components of the TA. (b) 80 MHz beat note height along with the measured value.

due to the frequency generator used for driving the AOM. We found that the sidebands arising due to the noise of the TA get embedded in the background with all the residue values we measured. Similar to the cavity signals, here also we try to correlate the height of the beat note at 80 MHz with the increasing residue [Fig. 7(b)]. This is again indirectly measured using inbuilt software in the spectrum analyzer. We find that the beat note height decreases with increasing residue, though with fluctuations present. Thus, our method and beat notes show the same trend qualitatively as far as the noise of the TA is concerned; the latter show fluctuations in the value at higher noise levels.

In the above results, we found that our method shows the same trend as is shown by the cavity and beat measurements, thus confirming our methodology. However, the variations in the latter two are smoother only for lower noise levels. As the residue increases, it shows the increase in noise and thus the fluctuations in the cavity mode, as well as the beat note height also increases. We think it is because of one advantage in our method. In the noise measurement, the resonant laser beam forms the background from which one needs to separate the noise part. In the case of cavity and beat, the ratio of signal (i.e., noise output of the TA) to the background is quite low, which results in higher fluctuation. However, in our method, we effectively cut the resonant beam using the vapor cell absorption. Thus, it becomes much easier to detect the noise output of the TA in absence of the resonant beam. Another point to note here is the effect of input power fluctuation/change in the case of cavity modes and beat note measurements. In our method, we have already noticed the absence of this effect for a wide range of power. One more point to emphasize in the case of the other two methods, due to the fluctuation in the output shown in Figs. 7(b) and 8(b), they cannot be used for optimizing the TA parameters as was done using our method in the previous section (Fig. 4) online. Height variation in the two cases was measured indirectly using a routine after analyzing the data. However, with the residue measurement, we could directly measure the residue. Hence, it can be used to readily optimize the working parameters of the TA, such as current and temperature online, while using

the TA in the experiment. This is because of a one-to-one relationship between these parameters and residue measured in our experiment. This is clearly shown in the data in Fig. 4. We further showed that the dynamic range of measurement in our method could be improved by heating the cell to allow the Doppler broadening to take care of the frequencies around the resonance part as well. This is shown in Fig. 5, where we have shown the temperature variation of the residue measurement. Thus, we could have a good measure of residue away from resonance as well.

5. CONCLUSIONS

In conclusion, we have devised a simple method to estimate the spectral characteristics and optimization of a TA output with regard to its noise component. We could use it to compare various TA chips used for different time periods in the experiments with a standard one (a brand new TA chip in our case). Further, our method gives a simple way to optimize the TA working parameters, including current and temperature online, compared to other traditional methods due to a one-to-one relationship between working parameters of TA and residue measured. Such a direct one-to-one relation was not found in other traditional methods. We optimized the working parameters of one of our chips, and the optimum value of the TA current was found to be around 1.2–1.4 A. We attribute this to the fact that at such current values, the spontaneous emission, and thus the ASE, becomes suppressed. This results in lower residue. Similarly, the residue is lower at a lower temperature of the TA chip. This is due to the fact that with lower temperature, the random spontaneous process gets suppressed and hence the ASE and the residue get reduced. The residue decreases with the seed power because of the frequency-pulling effect. We then briefly see the effect of temperature, magnetic inhomogeneity, collision effects, and transit time broadening. We see an increase in temperature increases the dynamic range of our method, and we could use it to measure the residue for higher input power. However, there is no effect of magnetic inhomogeneity, collision, or transit time in our conclusions because of a very low effect on shift/broadening of absorption

lines for our experimental parameters. We finally verify and compare our method with various spectral purity testing methods viz. spectrometer, FP cavity, and a heterodyne beat measurement. For the spectrometer, we find that the method has low resolution, so it cannot give any quantitative results. For the cavity mode and beat measurements, decreases in the cavity mode height and the beat note height are in sync with our results with increasing noise, thus verifying the conclusions of our method. However, as these methods give more qualitative and indirect results (by using a software routine), our method is better suited for optimization and characterization of the TA online. The advantage of our method over the other methods is the effective cutting down of the whole background of a resonant beam by using the vapor cell while measuring the residue. This makes the signal-to-noise ratio for our method much higher even at lower noise levels compared to other methods. It is noted here that noise in our case is the actual signal we are measuring. Further, our residue measurement can be better in the case of any input power fluctuation or change when compared to the cavity/beat measurements for a wide range of input powers, as pointed out before. This range could be further improved by heating the cell, which clearly shows the increase in the dynamic range of our method due to higher absorption at higher temperature.

Thus, our method gives a simple parameter to measure the noise of a TA-based laser system using a multi-pass vapor cell in the mid-IR region. This wavelength is important for various laser-cooling experiments as well as in communication due to its proximity with second harmonic of the important C band. Advantages using these cells are their availability in a typical atomic physics laboratory and their easy-to-fabricate design. It can be a handy parameter for optimization of a TA. A one-to-one relationship between the residue measured and TA parameter gives our method an edge with regard to its suitability for optimization when compared to other traditional methods. The results are in sync with traditional methods used for it viz. cavity modes and beat note measurements, which can give only qualitative results for TA noise. Our measurement can be used to suppress the noise and to improve precision in atom interferometers, atomic clocks, and other atomic manipulations. Further, it gives a simple parameter to study the non-linearity of the TA chip, which can be used to understand phenomena such as four wave-mixing and terahertz generation better. It can also be used to study ASE to explore its future use, including bioimaging using optical coherent tomography and far-off-resonance atom traps.

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